

GEOL 211 • Exam 2 — Review Guide Answers

Every review-guide bullet, in order, answered from the class slides. Red = the key part of the answer. Also review your class assignments.

ATMOSPHERIC CIRCULATION

1 Explain how/why air density changes.

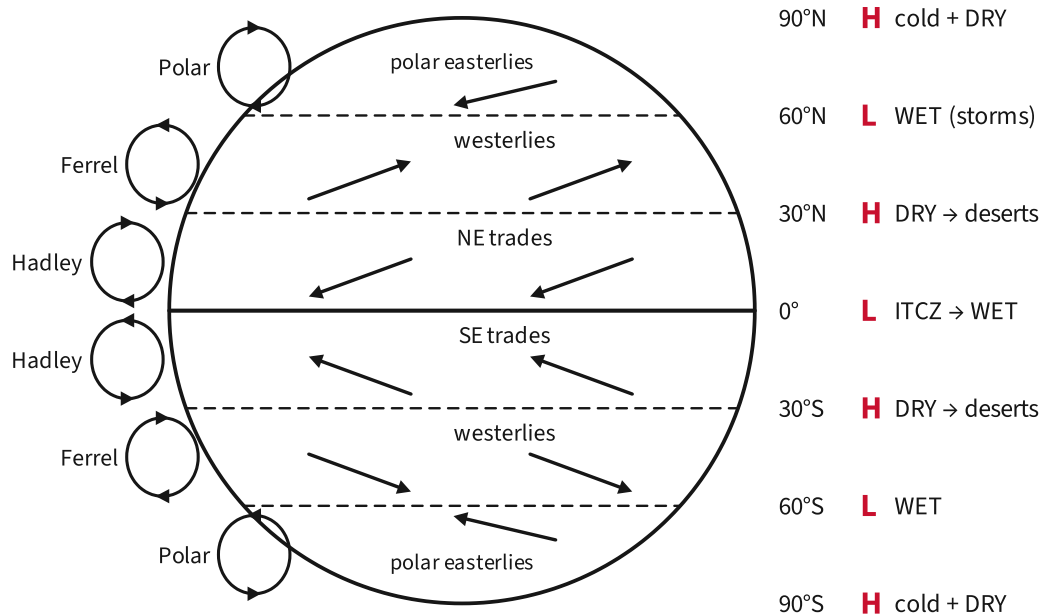
- Air density is controlled by **temperature** and **moisture content**.
- Warm air: molecules spread out → **less dense → rises**. Cold air: molecules packed together → **denser → sinks**.
- Water vapor (H_2O) is lighter than the N_2 (78% of air) and O_2 (21%) it replaces → **humid air is LESS dense than dry air**.
- Densest air on Earth = the **poles** (cold + dry).

2 Explain why regions of high/low density develop. Identify areas of high/low pressure. Describe the weather with each.

- The Sun heats Earth **unevenly** (differential heating): the equator gets direct sunlight and is hot; the poles get spread-out sunlight and are cold.
- Warm air rises → **LOW pressure**: rising air cools, water vapor condenses → **clouds, rain, storms**.
- Cold air sinks → **HIGH pressure**: sinking air warms → **clear, dry** weather.
- Air flows from **high → low** pressure = wind.
- Where: **LOW at 0° and 60°** · **HIGH at 30° and 90°**.

3 Draw the global atmospheric circulation pattern. Label high/low pressure, the ITCZ, and high/low precipitation.

- 3 cells per hemisphere: **Hadley** (0–30°), **Ferrel** (30–60°), **Polar** (60–90°).
- **0° = LOW = ITCZ** (trade winds meet, air rises) → **heavy rain**. 30° = HIGH → **dry, deserts**. 60° = LOW → **wet**. 90° = HIGH → **cold, dry**.
- Surface winds: **trade winds** (0–30°), **westerlies** (30–60°), **polar easterlies** (60–90°).



Loops = cells. Arrows = surface winds, from H toward L, bent right in the NH and left in the SH.

4 Explain why the Coriolis Force exists and what the result is.

- **Why:** Earth rotates. The equator travels farther each day (~40,000 km) than higher latitudes (~20,000 km at 60°), so it moves faster. A moving object keeps its original speed, so its path appears to curve.
- **Result:** objects are deflected **to the RIGHT in the Northern Hemisphere and to the LEFT in the Southern Hemisphere**.
- No Coriolis at the equator; it increases with latitude, speed and mass; it only matters for large motions (tens of km or more).
- It does **not** cause wind — it changes the wind's **direction**.

5 Explain the development of monsoons.

- Land heats up and cools down **faster than water**.
- **Summer:** land is hotter → air rises over land (LOW) → moist ocean air flows onto land → **heavy rain (wet season)**.
- **Winter:** land is colder → air sinks over land (HIGH) → dry air flows from land to sea → **dry season**.
- Monsoons follow the ITCZ: June–July it's north over India (rain there); Dec–Jan it's south (rain in Darwin, Australia).

6 Explain the development of land/sea breeze.

- Same cause as monsoons (land heats/cools faster than water), but on a **daily** cycle.
- **Day:** land warmer → air rises over land → cool air flows from sea to land = **SEA BREEZE**.
- **Night:** land cooler → air sinks over land → air flows from land to sea = **LAND BREEZE**.

7 List and describe the conditions necessary for a major hurricane to develop.

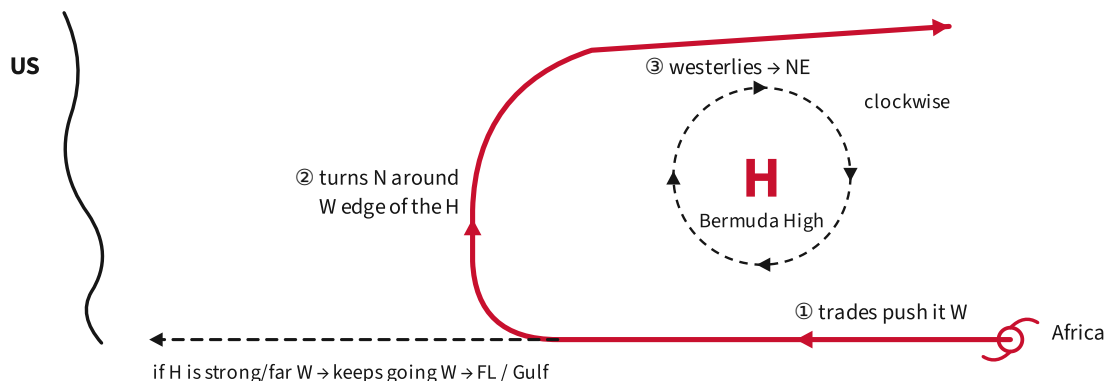
- **Warm water ($>27^{\circ}\text{C}$)** — supplies the heat and moisture (fuel).
- **Low-pressure system** — air flows in and rises.
- **Warm air with high water vapor** — when vapor condenses it releases **latent heat**, which powers the storm.
- **Low wind shear (<20 knots)** — keeps the storm upright; strong shear tears it apart.
- **Strong Coriolis** — makes it spin (cyclonic flow: counterclockwise in NH), so no hurricanes at the equator.

8 List and describe the risks associated with hurricanes.

- Hazards: **waves, wind, rain, flooding, landslides, storm surge**. Rain flooding can reach far inland (Ivan $\rightarrow$ Asheville NC, 2004).
- **Storm surge** is caused by strong **onshore winds** pushing water onto the coast; worst at high tide (Sandy 2012).
- Surge is higher with: stronger **wind speed** · bigger **storm radius** · **forward speed** (fast = higher surge, slow = longer flooding) · **perpendicular** (head-on) approach · **wide, gently sloping continental shelf** · **coastline shape** that funnels water.

9 Predict the path of a hurricane based on the location of high/low pressure systems.

- Air flows from high to low, so hurricanes are **pushed around HIGH pressure** (clockwise around a NH high) and **pulled toward LOW pressure**.
- Atlantic: trade winds push it **west** $\rightarrow$ it curves **north** around the west side of the Bermuda High $\rightarrow$ westerlies push it **northeast**.
- High extends far west $\rightarrow$ storm keeps going west (Florida/Gulf). High farther east $\rightarrow$ storm turns north sooner (East Coast or out to sea).



10 Explain expected changes in hurricane activity due to climate change and trends observed over the last 100 years.

- **Expected:** warmer water = more fuel $\rightarrow$ **stronger, wetter storms**; higher sea level $\rightarrow$ **higher storm surge**. Not necessarily more storms.
- **Last 100 years:** no clear trend in the **number** of hurricanes (older records missed storms), but a larger share of **strong** (Cat 3–5) storms, more rain, and more damage as coasts are built up.

OCEAN CIRCULATION

11 Describe the properties of surface and deep water.

	Surface layer	Deep layer
Temperature	warm	cold (~4°C)
Density	low	high
Moved by	wind · 10% of ocean · fast	density/gravity · 90% · slow
Mixing	mixed by wind	not mixed by wind
Nutrients	low	high

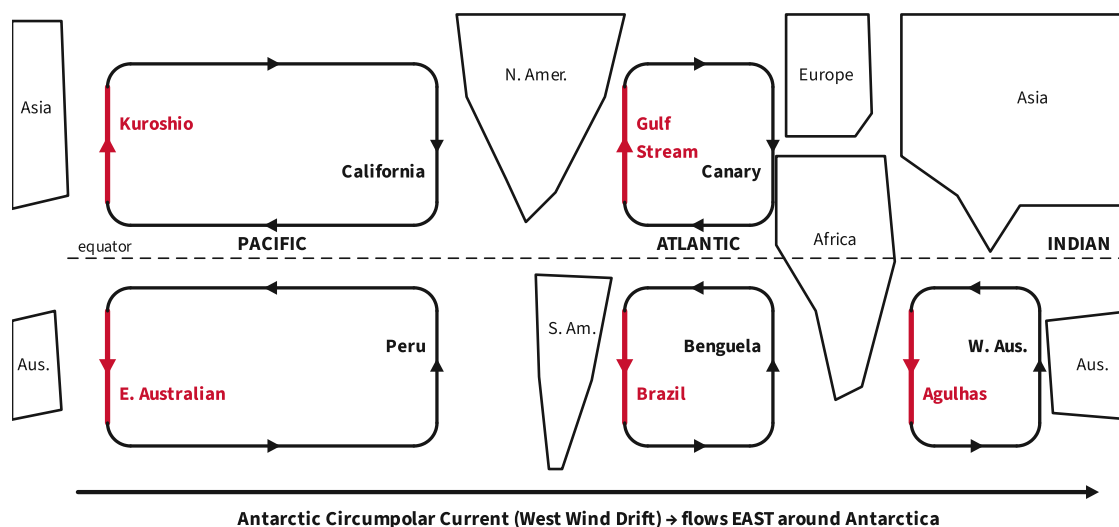
- Between them is the **thermocline**: temperature drops fast from ~13°C at 200 m to ~4°C at 1000 m, so the layers don't mix.
- Salinity ≈ 35 ppt; highest in the subtropics (23.5–35°, evaporation) and lowest near rivers and rainy areas.

12 Recognize the major oceanic currents.

- Currents form **gyres**: **clockwise in the NH**, **counterclockwise in the SH**. The **west** side of each gyre is warm; the **east** side is cold.

Gyre	West side (warm)	East side (cold)
N Atlantic	Gulf Stream	Canary
S Atlantic	Brazil	Benguela
N Pacific	Kuroshio	California
S Pacific	East Australian	Peru
Indian	Agulhas	West Australian

- Also: Antarctic Circumpolar Current flows east around Antarctica.



13 Describe the forces causing surface and deep currents.

- Ocean circulation is controlled by **wind, density, the position of the continents, and Coriolis**.
- **Surface currents** = **driven by WIND** (10% of ocean, fast). Coriolis turns them: **Ekman transport** moves water **90° to the right of the wind** in the NH (left in SH).
- **Deep currents** (thermohaline) = **driven by DENSITY** (temperature + salinity) and gravity (90% of ocean, slow).
- Density is set at the surface: **temperature** (cooling by the air) and **salinity** (evaporation or sea-ice formation raise it; rain lowers it). Cold + salty water sinks.

14 Explain why the speed of surface currents varies (eastern/western boundary currents).

- Western boundary currents (Gulf Stream) are **fast, narrow, deep, warm**. Eastern boundary currents (Canary) are **slow, wide, shallow, cold**.
- **Why**: the **eastward rotation of Earth** and **Coriolis increasing with distance from the equator** compress the western boundary current, so its speed increases.

15 Explain how surface currents affect weather.

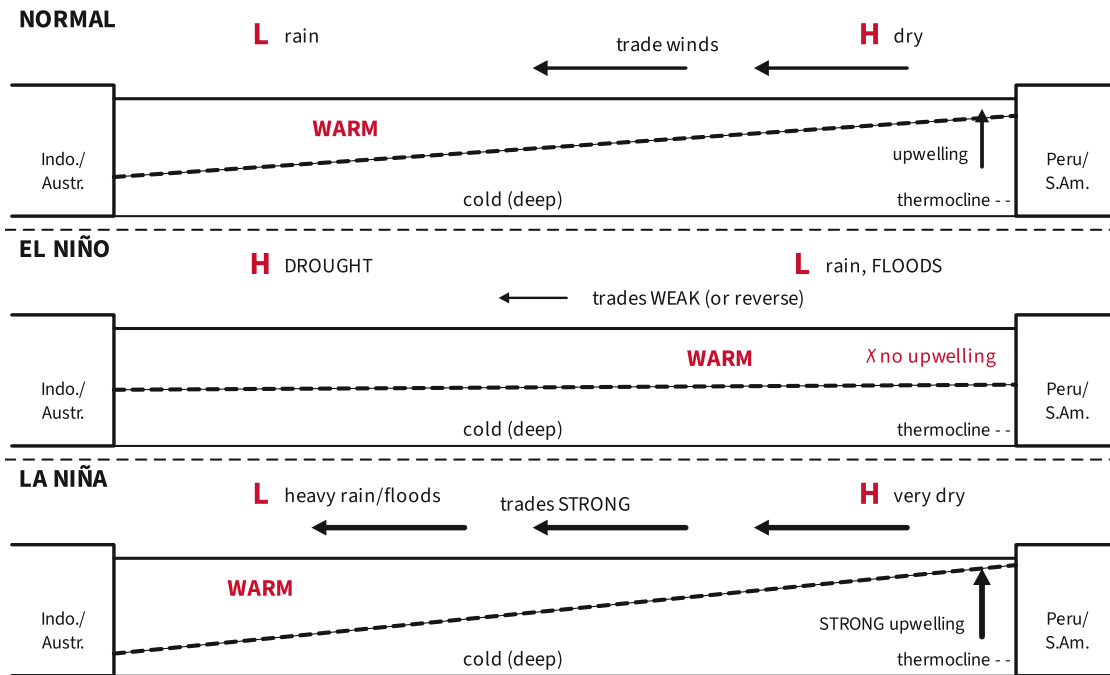
- Currents **transport heat** from the equator toward the poles.
- Warm currents warm the air → **milder, wetter climates**. Ex: Gulf Stream keeps Europe warm — January avg: **Newfoundland 30/16°F vs Ireland 46/37°F** (same latitude).
- Cold currents cool the air → cooler, drier coasts.

16 Explain how deep and surface water mix (upwelling/downwelling). Implications for marine life and climate?

- **Upwelling** = deep, cold, nutrient-rich water rises to the surface. **Downwelling** = surface water sinks.
- **Coastal**: wind along the coast + Ekman transport pushes surface water **offshore** → deep water rises (upwelling); pushed **onshore** → water piles up and sinks (downwelling).
- **Equatorial** upwelling (water moves apart at the equator) and **Southern Ocean** upwelling (around Antarctica).
- **Marine life**: upwelling brings **nutrients** to the sunlit surface → phytoplankton → rich fisheries. Downwelling carries **oxygen** down to deep water.
- **Climate**: mixing moves heat and gases (O₂, CO₂) between the surface and deep ocean; upwelled cold water cools the coast.

17 Recognize and explain the causes/effects of El Niño/La Niña. How does this affect hurricane activity?

- **Normal**: trade winds push warm water west → warm and rainy in Indonesia/Australia; upwelling of cold water off Peru.
- **El Niño**: **trade winds weaken** → warm water moves east → upwelling off Peru stops (fish decline); rain/floods in Peru, drought in Australia.
- **La Niña**: **trade winds strengthen** → even more warm water west, colder east Pacific, stronger upwelling.
- **Hurricanes**: El Niño → **fewer Atlantic hurricanes** (more wind shear), more Pacific. La Niña → **more Atlantic hurricanes** (less shear), fewer Pacific.
- Anomaly = difference from the average (how SST maps show El Niño/La Niña).



Dashed line = thermocline. Normal = tilted, El Niño = flat, La Niña = very tilted.

18 Explain how deep water circulation affects climate.

- Deep water forms near the poles where water is **cold and salty** (sea ice leaves salt behind) → dense → sinks: **North Atlantic Deep Water** (near Greenland) and **Antarctic Bottom Water**.
- This drives the **conveyor belt**: warm surface water flows to the North Atlantic, sinks, travels through the deep Atlantic, Indian and Pacific, rises, and returns.
- It **transports heat** toward the poles (keeps Europe mild) and stores gases (CO₂, O₂) in the deep ocean.
- If it slows (fresh meltwater makes water less dense so it can't sink) → **abrupt cooling**. Ex: **Younger Dryas** — Lake Agassiz meltwater → near-glacial cold; it ended with Greenland warming 10°C in a decade. The Atlantic overturning slowed in the 20th century.